THE HAFNIAN VAN DER WAERDEN CONJECTURE FAILS IN ORDER SIX

ABSTRACT. Friedland formulated a hafnian analogue of the van der Waerden–Tverberg formula and reported that Gurvits had conjectured its perfect-matching case. Friedland also records that this case is false once n is sufficiently large, an observation he credits to S. Norin. This note locates the least order at which failure occurs: it is $n = 3$, on six vertices. For an explicit matrix $B \in \Psi_6$,

$$\operatorname{haf}(B) = \frac{29}{243} < \frac{3}{25} = \operatorname{haf}\left(\frac{1}{5}A(K_6)\right),$$

so the conjectured minimum fails at $n = 3$. Failure cannot occur earlier: the conjecture holds at $n = 2$, a case subsumed by Friedland's Lemma 3.1, for which we include a short direct argument. What is new here is the explicit order-six matrix and the minimal order it fixes, not the failure of the conjecture itself. We do not determine the exact minimum on Ψ_6.

For a perfect matching M of K_{2n}, let $A(M)$ denote its symmetric adjacency matrix, and set

$$\Psi_{2n} := \operatorname{conv}\{A(M) : M \text{ is a perfect matching of } K_{2n}\}.$$

This is the set called Ψ_{2n} in [2, 3], where it is introduced as the convex set of symmetric doubly stochastic matrices with zero diagonal whose extreme points are the symmetric permutation matrices with zero diagonal, and is then described by Edmonds' odd-set inequalities [2, Eq. (1.2)]. Those inequalities are part of the description, so the domain of the conjecture is the perfect matching polytope displayed above, and not the set cut out by symmetry, nonnegativity, zero diagonal and double stochasticity alone. For a symmetric matrix $C = [c_{ij}] \in \mathbb{R}_{\geq 0}^{2n\times 2n}$ with zero diagonal, define

$$\operatorname{haf}(C) := \sum_{M} \prod_{\{i,j\}\in M} c_{ij},$$

where the sum is over the perfect matchings of K_{2n}, and let

$$\mu_n := \min_{C\in\Psi_{2n}} \operatorname{haf}(C).$$

Friedland formulated the corresponding family of hafnian conjectures and reported that Gurvits had stated the perfect-matching case [2, Eq. (1.4) and the discussion following it]; see also [3, Eq. (6.3)]. That family is indexed by the size k of the matchings counted, and the perfect-matching case is $k =$

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n. In the present notation, the perfect-matching hafnian van der Waerden conjecture asserts

$$\mu_n = \operatorname{haf}\left(\frac{1}{2n-1}A(K_{2n})\right), \qquad n \geq 2. \tag{1}$$

That (1) is false for large n is stated outright in both printings: "for $k = n$ and n big enough (1.4) is wrong" [2, following Eq. (1.4)], and likewise for (6.3) in [3, §6]. Friedland thanks S. Norin for pointing out that failure, and the argument recorded there proceeds through a family of graphs of Cygan, Pilipczuk and Škrekovski [1]; neither that assertion nor its proof is claimed here. What follows instead is the least order at which failure occurs, together with an explicit matrix that witnesses it.

Theorem 1. *Let*

$$B = \frac{1}{9}\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 1 \\ 3 & 0 & 3 & 1 & 1 & 1 \\ 3 & 3 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & 1 & 3 & 3 & 0 \end{pmatrix}.$$

Then $B \in \Psi_6$ and

$$\operatorname{haf}(B) = \frac{29}{243} < \frac{3}{25} = \operatorname{haf}\left(\frac{1}{5}A(K_6)\right).$$

Consequently, (1) *fails at $n = 3$. It holds at $n = 2$, a case subsumed by* [2, Lemma 3.1] *and reproved directly below, so $n = 3$ is the least admissible value for which it fails.*

Proof. Partition $[6]$ into

$$T_1 = \{1, 2, 3\}, \qquad T_2 = \{4, 5, 6\}.$$

For $i \in T_1$ and $j \in T_2$, define the perfect matching

$$M_{ij} := \{\{i, j\}, T_1 \setminus \{i\}, T_2 \setminus \{j\}\};$$

the last two sets have cardinality two and hence represent edges. Every edge between T_1 and T_2 occurs in exactly one M_{ij}, whereas every edge inside either triangle occurs in exactly three of them. Therefore

$$B = \frac{1}{9}\sum_{i \in T_1}\sum_{j \in T_2} A(M_{ij}),$$

so $B \in \Psi_6$.

Every perfect matching of K_6 uses an odd number of edges between T_1 and T_2, hence either one or three. There are nine matchings with one such edge, each of weight

$$\frac{1}{9}\left(\frac{1}{3}\right)^2 = \frac{1}{81},$$

and six matchings with three such edges, each of weight $1/9^3$. Thus

$$\operatorname{haf}(B) = 9\left(\frac{1}{81}\right) + 6\left(\frac{1}{729}\right) = \frac{29}{243}.$$

Since K_6 has $5!! = 15$ perfect matchings,

$$\operatorname{haf}\left(\frac{1}{5}A(K_6)\right) = \frac{15}{5^3} = \frac{3}{25},$$

and

$$\frac{3}{25} - \frac{29}{243} = \frac{4}{6075} > 0.$$

It remains to verify minimality of the order. Let M_1, M_2, M_3 be the three perfect matchings of K_4. Every $C \in \Psi_4$ has the form

$$C = \alpha A(M_1) + \beta A(M_2) + \gamma A(M_3), \qquad \alpha, \beta, \gamma \geq 0, \qquad \alpha + \beta + \gamma = 1.$$

The three matchings partition the edges of K_4, and therefore

$$\operatorname{haf}(C) = \alpha^2 + \beta^2 + \gamma^2 \geq \frac{(\alpha+\beta+\gamma)^2}{3} = \frac{1}{3}.$$

Equality holds precisely at $\alpha = \beta = \gamma = 1/3$, which gives $C = \frac{1}{3}A(K_4)$ and

$$\operatorname{haf}\left(\frac{1}{3}A(K_4)\right) = 3\left(\frac{1}{3}\right)^2 = \frac{1}{3};$$

so $\mu_2 = \frac{1}{3}$, attained at $\frac{1}{3}A(K_4)$ and nowhere else. Hence (1) holds at $n = 2$; this is subsumed by the case $k = 2$ of [2, Lemma 3.1], and the short direct argument is included only to keep the minimality half of the theorem self-contained. □

Remark 2 (Consistency with local minimality). Friedland proved that $(2n-1)^{-1}A(K_{2n})$ is a strict local minimizer of the hafnian on Ψ_{2n} [2, Lemma 3.1]. Theorem 1 shows that it is already not a global minimizer at $n = 3$. The two are compatible because B is nowhere near $\frac{1}{5}A(K_6)$. Indeed a perfect matching of K_6 has at most one edge inside the triple $T_1 = \{1,2,3\}$, so every $C = [c_{ij}] \in \Psi_6$ satisfies $\sum_{\{i,j\}\subseteq T_1} c_{ij} \leq 1$; this is tight at B, whose entries inside T_1 sum to 1, and strict at $\frac{1}{5}A(K_6)$, where they sum to $3/5$. So B lies on the boundary of Ψ_6, in a supporting hyperplane that $\frac{1}{5}A(K_6)$ misses. Quantitatively, set $C_t := (1-t)\frac{1}{5}A(K_6) + tB$ for $t \in [0,1]$. Each C_t again has constant entries $(3+2t)/15$ inside the two triples and $(9-4t)/45$ between them, so the same $9+6$ decomposition gives

$$\begin{aligned}\operatorname{haf}(C_t) &= \frac{9(9-4t)(3+2t)^2}{45 \cdot 15^2} + \frac{6(9-4t)^3}{45^3} \\ &= \frac{729 + 108t^2 - 112t^3}{6075} = \frac{3}{25} + \frac{4t^2(27-28t)}{6075}.\end{aligned}$$

Thus $\operatorname{haf}(C_t) > \frac{3}{25}$ for $0 < t < \frac{27}{28}$, with equality at $t = \frac{27}{28}$: the hafnian rises out of $\frac{1}{5}A(K_6)$, peaks at $t = \frac{9}{14}$, and drops below the conjectured value only on the final $\frac{1}{28}$ of the segment.

Remark 3 (Scope and prior work). Theorem 1 refutes (1) at $n = 3$ by an explicit witness and fixes $n = 3$ as the least order at which a witness can exist. It is not the first refutation of (1): both [2] and [3] state that the conjecture is wrong for $k = n$ with n large, and no novelty is claimed for that assertion or for the asymptotic argument behind it. Nor is the case $n = 2$ new; it is subsumed by the case $k = 2$ of [2, Lemma 3.1]. What is offered here is the matrix B, the exact value $\mathrm{haf}(B) = 29/243$, and the resulting identification of $n = 3$ as the minimal order. Theorem 1 gives only $\mu_3 \leq 29/243$; no equality claim is made, and the status of the other members of Friedland's family of hafnian conjectures is not addressed.

References

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